



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computing

Semester 01 2025/2026

Subject : Database Programming (SECP3623)
Section : Section 1-2
Task : Project I
Due : 28th November 2025 (10%)

Project Overview

This mini project is to enable students to **design and implement the backend logic of a database application system** using **MySQL**. Students are expected to show **the database operations are developed through SQL statements**.

Note: This project focuses on SQL backend development only. No front-end or user interface implementation is required.

Through this project, students will:

1. Apply the full range of **SQL statements** learned (DDL, DML, DQL).
2. Demonstrate **data processing logic** using filtering, aggregation, subqueries, joins, and functions.
3. Implement **indexing** for query optimization and explain performance.
4. Work collaboratively to combine individual SQL tasks into a single functional system.
5. Communicate results effectively through a structured **report** and a demonstration presentation must be included. You may provide the YouTube link or Video URL of your group presentation within the report. And the link must be accessible and viewable (workable) without login restrictions.

By the end of this project, each group (3-4 members) should be able to demonstrate a **working backend** capable of supporting common system operations (for example: reservation, sales, record management, attendance, or student system) through SQL programming.

Project Requirements

A. Database Creation (DDL)

- Create database and tables using `CREATE DATABASE` and `CREATE TABLE`.
- Apply constraints such as `PRIMARY KEY`, `FOREIGN KEY`, `NOT NULL`, `UNIQUE`, `CHECK`, and `DEFAULT`.
- Include `ALTER TABLE` and `DROP TABLE` statement.

B. Data Manipulation (DML)

- Insert **at least 10 records per table** using INSERT.
- Perform updates and deletions with appropriate conditions (UPDATE, DELETE with WHERE).

C. Data Retrieval (DQL / SELECT)

Develop at least **following SQL queries** that demonstrate:

1. Filtering (WHERE, AND, OR, NOT, BETWEEN, LIKE, NULL) .
2. Sorting (ORDER BY, LIMIT).
3. Aggregation (COUNT, SUM, AVG, MAX, MIN).
4. Grouping and filtering groups (GROUP BY, HAVING).
5. Numeric and string functions (ROUND, TRUNCATE, UPPER, LENGTH, CONCAT, SUBSTR).
6. Conditional logic (CASE WHEN).
7. Subqueries (single-row, multiple-row, correlated).
8. Set operations (UNION, NOT EXISTS).
9. Joins (NATURAL, INNER, LEFT OUTER, SELF).

D. Indexing and Optimization

- Create **two indexes**:
 - **BTREE** (for numeric/date columns).
 - **TEXT** (for text search).
- Analyse query plans before and after index creation.
- Write a short interpretation of what changed (rows examined, possible keys).

Project Report

Report should be clear, concise, and technically detailed. Below is the required structure:

Section	To Include
Introduction	● Brief background of the chosen domain. ● Objectives of the system and database backend. ● List of team members and their assigned roles. ● Short explanation of what each .sql file does.
SQL Implementation Summary	● Example snippets of important SQL statements (CREATE TABLE, INSERT, SELECT, INDEX). ● Mention constraints used and why. ● Screenshots of query execution in MySQL Workbench.
Query Demonstrations	● Results table + short explanation of what the query proves (grouping by category, filtering by condition). ● Representative query results.
Optimization Section	● Show your result before and after indexing. ● Briefly explain changes in key usage.
Team Work	● Each member describes their role and what they learned.
Conclusion	● Challenges faced and suggestions for improvement.

Tips for report writing:

- Use screenshots with labels and captions.
- Do not paste all SQL codes include only representative examples.
- Ensure consistent formatting and clear headings.

Presentation Requirements

- **Duration:** 10 minutes per group.
- **Tools:** MySQL Workbench + PowerPoint.
- **Presentation flow:**
 1. Brief introduction of system and objective.
 2. Show key SQL functionalities (DDL, DML, SELECT examples).
 3. Demonstrate indexing improvement.
 4. Team roles (Every individual shall present).

Submission Items

- Project Report + Presentation URL
- Presentation Slides
- Zipped folder of .sql files

Marking Rubric

SQL Implementation (CLO1)

Criteria	Excellent (5)	Good (4)	Fair (3)	Weak (2)	Poor (1)	Marks
Backend SQL Implementation (DDL/DML/Queries)	All SQL statements execute flawlessly; covers full range – DDL, DML, SELECT, joins, subqueries, functions.	Most SQL correct; small syntax or logic issues.	Several working statements but limited variety.	Few working statements; repetitive or basic logic.	Non-functional / incomplete SQL.	
Advanced Data Logic (Filtering, Aggregation, Subquery)	Uses complex logic (nested queries, GROUP BY, HAVING, CASE, conditions) effectively.	Correct use of aggregations with minor gaps.	Some queries too simple or misapplied.	Few aggregations or logical errors.	Not attempted.	
Indexing & Optimization	Correctly creates BTREE and TEXT indexes; shows analysis before/after; interprets performance gain.	Indexes implemented; partial analysis.	Indexing attempted without analysis.	Incomplete index usage.	Not attempted.	
Interpretation & Justification	Demonstrates strong understanding of how SQL logic supports system workflow; clearly explains data relationships, query results, and design rationale.	Shows good understanding with minor gaps.	Basic interpretation of results without linking to system logic.	Minimal explanation; unclear reasoning.	No reflection or incorrect interpretation.	
Subtotal						

Teamwork and Collaboration (CLO3)

Criteria	Excellent (5)	Good (4)	Fair (3)	Weak (2)	Poor (1)	Marks
Team Coordination & Contribution	Clear task division; balanced participation; strong collaboration evident in report and presentation; each member contributes to SQL work.	Good teamwork and participation with minor imbalance.	Some collaboration; limited task clarity.	Minimal teamwork; dominated by few members.	No teamwork evidence; incomplete group submission.	
Subtotal						

Documentation & Presentation (CLO4)

Criteria	Excellent (5)	Good (4)	Fair (3)	Weak (2)	Poor (1)	Marks
Report Documentation	Clear and complete report including intro, SQL summary, query screenshots, optimization, and reflection.	Report complete with minor formatting or clarity issues.	Adequate report but lacks details or screenshots.	Weakly structured or incomplete sections.	Missing or very poor documentation.	
Slide Quality & Visual Communication	Slides are professional, organised, visually clear, and consistent with project flow.	Well-prepared slides with minor design issues.	Slides understandable but inconsistent layout.	Poorly structured or cluttered slides.	Missing or unreadable slides.	
Verbal Presentation	Excellent delivery, confident speakers, balanced time usage, fluent explanation, and effective Q&A handling.	Good delivery; some coordination issues.	Understandable but low energy; moderate Q&A.	Disorganized delivery or weak explanations.	Unable to present or answer questions.	
Subtotal						

